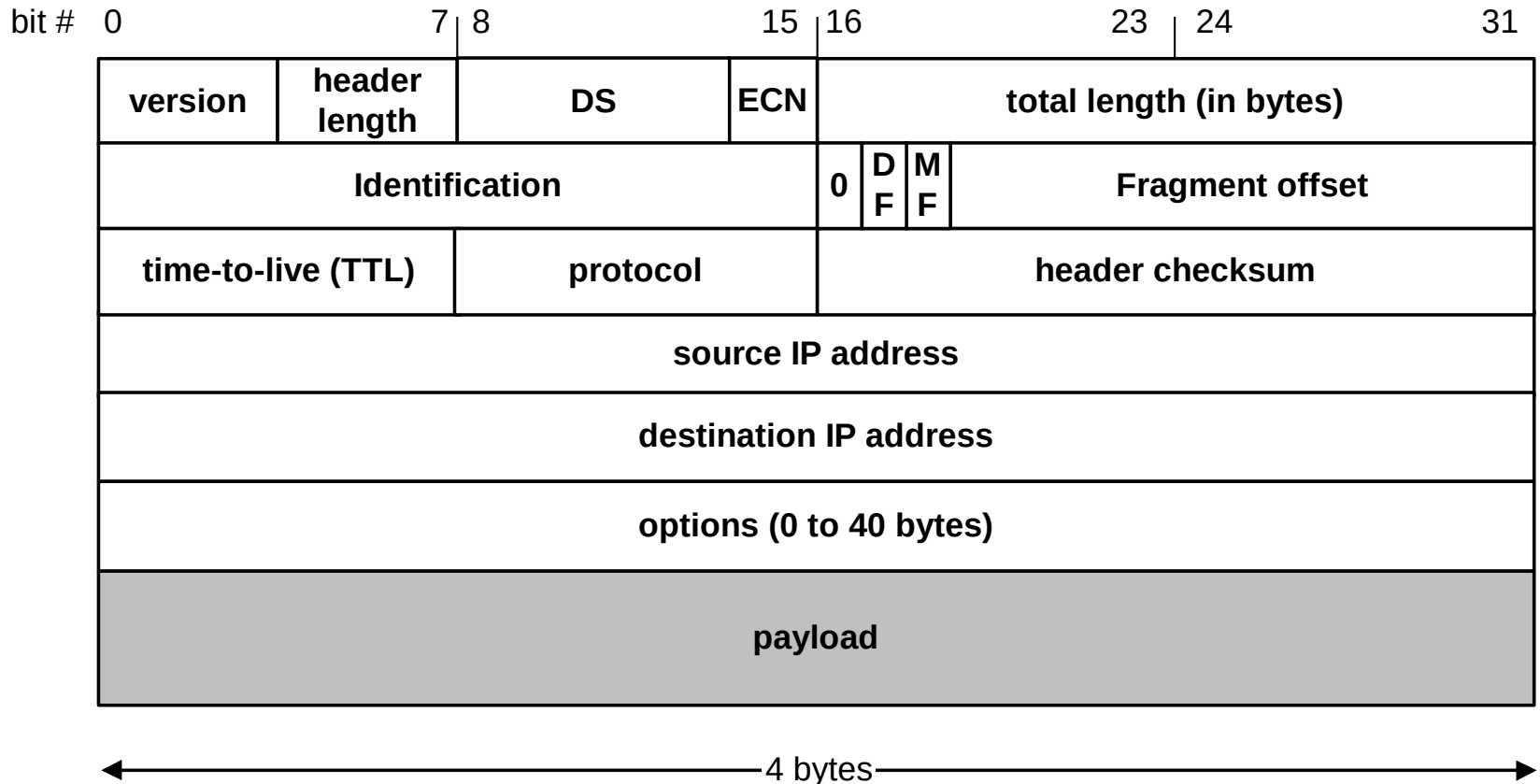


IPv4 Header

IP Datagram Format



- 20 bytes ≤ Header Size < 60 bytes
- 20 bytes ≤ Total Length < 65536 bytes

Internet Protocol Version 4, Src: 192.168.82.147 (192.168.82.147), Dst: 192.243.232.2 (192.243.232.2)

Version: 4

Header Length: 20 bytes

Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))

0000 00.. = Differentiated Services Codepoint: Default (0x00)

.... ..00 = Explicit Congestion Notification: Not-ECT (Not ECN-Capable Transport) (0x00)

Total Length: 1155

Identification: 0x69de (27102)

Flags: 0x02 (Don't Fragment)

0... = Reserved bit: Not set

.1... = Don't fragment: Set

..0. = More fragments: Not set

Fragment offset: 0

Time to live: 128

Protocol: TCP (6)

Header checksum: 0xd064 [validation disabled]

[Good: False]

[Bad: False]

Source: 192.168.82.147 (192.168.82.147)

Destination: 192.243.232.2 (192.243.232.2)

[Source GeoIP: Unknown]

[Destination GeoIP: Unknown]

Transmission Control Protocol, Src Port: 57487 (57487), Dst Port: 80 (80), Seq: 1102, Ack: 883, Len: 1115

- Transmission is row by row
- For each row:
 1. First transmit bits 0-7
 2. Then transmit bits 8-15
 3. Then transmit bits 16-23
 4. Then transmit bits 24-31

Fields of the IP Header

- **Version (4 bits):** current most widely used version is 4.
- **Header length (4 bits):** length of IP header, in multiples of 4 bytes
- **DS/ECN field (1 byte)**
 - This field was previously called as Type-of-Service (TOS) field. The role of this field has been re-defined, but is “backward compatible” to TOS interpretation
 - **Differentiated Service (DS) (6 bits):**
 - Used to specify service level
 - **Explicit Congestion Notification (ECN) (2 bits):**
 - Feedback mechanism used by TCP

Fields of the IP Header

- **Identification (16 bits):** Unique identification number for each datagram set at host end.
- **Flags (3 bits):**
 - First bit always set to 0
 - DF bit (Do not fragment)
 - MF bit (More fragments)

Fields of the IP Header

- **Time To Live (TTL) (1 byte):**

- Specifies longest paths before datagram is dropped
- Role of TTL field: Ensure that packet is eventually dropped when a routing loop occurs

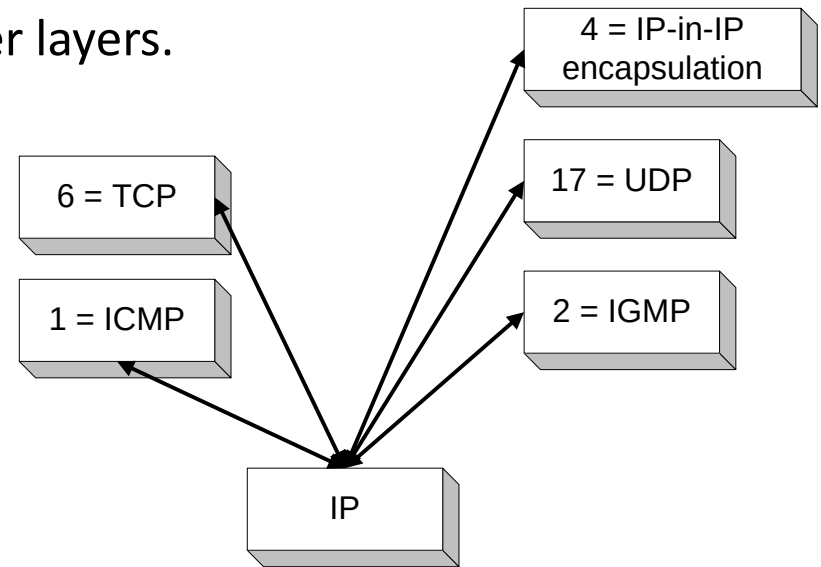
Used as follows:

- Sender sets the value (e.g., 64)
- Each router decrements the value by 1
- When the value reaches 0, the datagram is dropped

Fields of the IP Header

- **Protocol (1 byte):**

- Specifies the higher-layer protocol.
- Used for de-multiplexing to higher layers.



- **Header checksum (2 bytes):** The IP checksum is a 16 bit 1's complement sum of all the 16 bit words in the IP header.

Fields of the IP Header

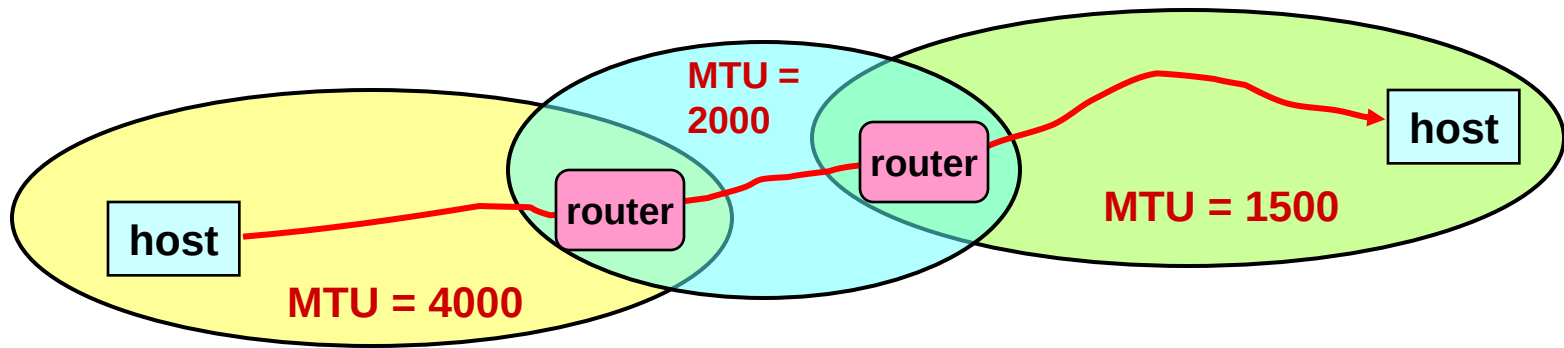
- **Options:**
 - Record Route: each router that processes the packet adds its IP address to the header.
 - Timestamp: each router that processes the packet adds its IP address and time to the header.
 - Source Routing: specifies a list of routers that must be traversed.
- **Padding:** Padding bytes are added to ensure that header ends on a 4-byte boundary

Maximum Transmission Unit

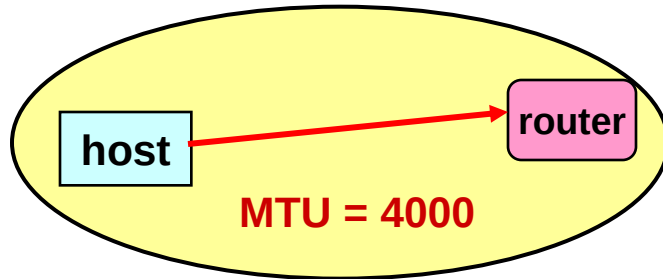
- Maximum size of IP datagram is 65535, but the data link layer protocol generally imposes a limit that is much smaller
- The limit on the maximum IP datagram size, imposed by the data link protocol is called **maximum transmission unit (MTU)**
- MTUs for various data link layers:

Ethernet:	1500
PPP:	296

IP Fragmentation



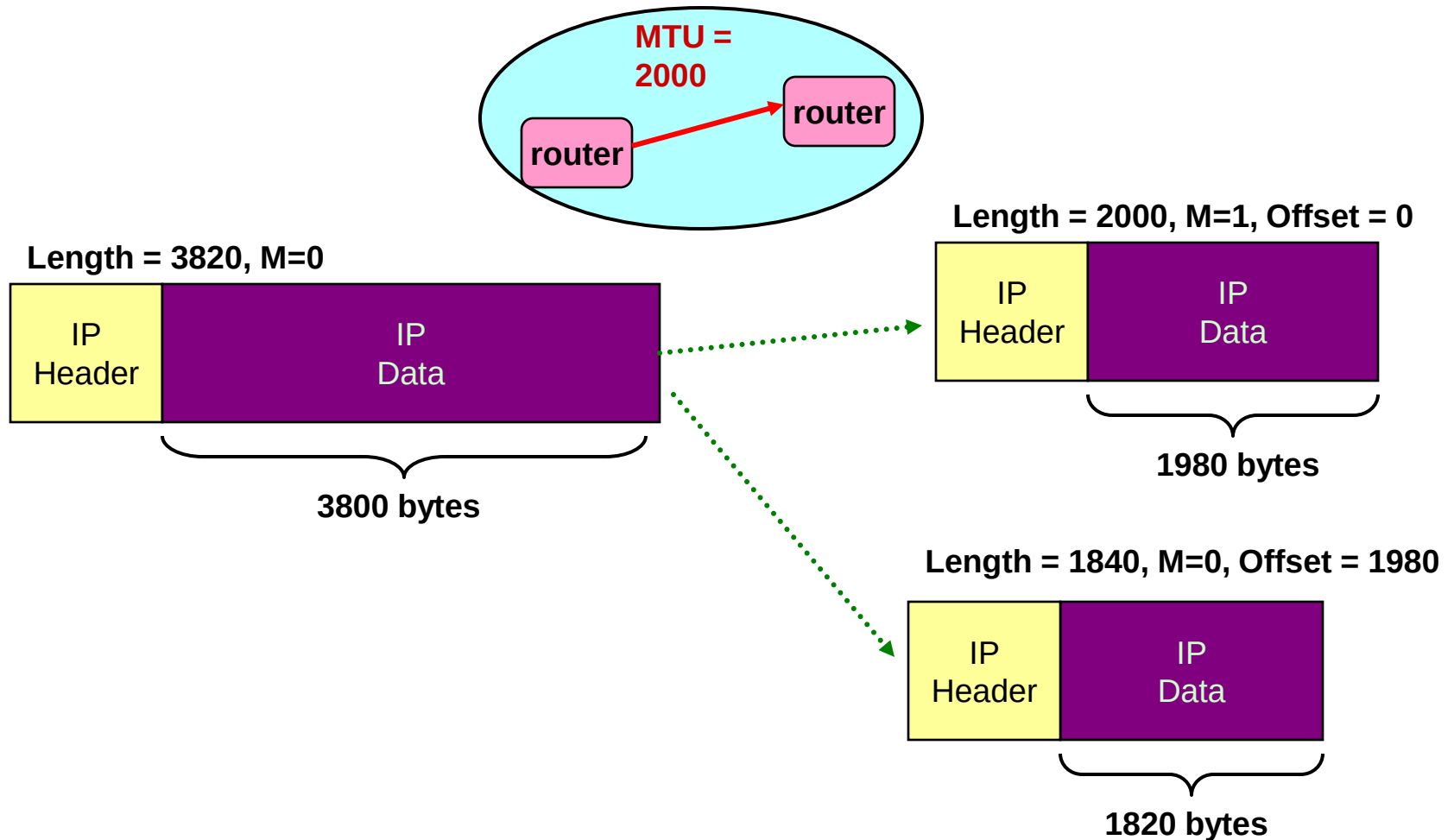
IP Fragmentation Step #1



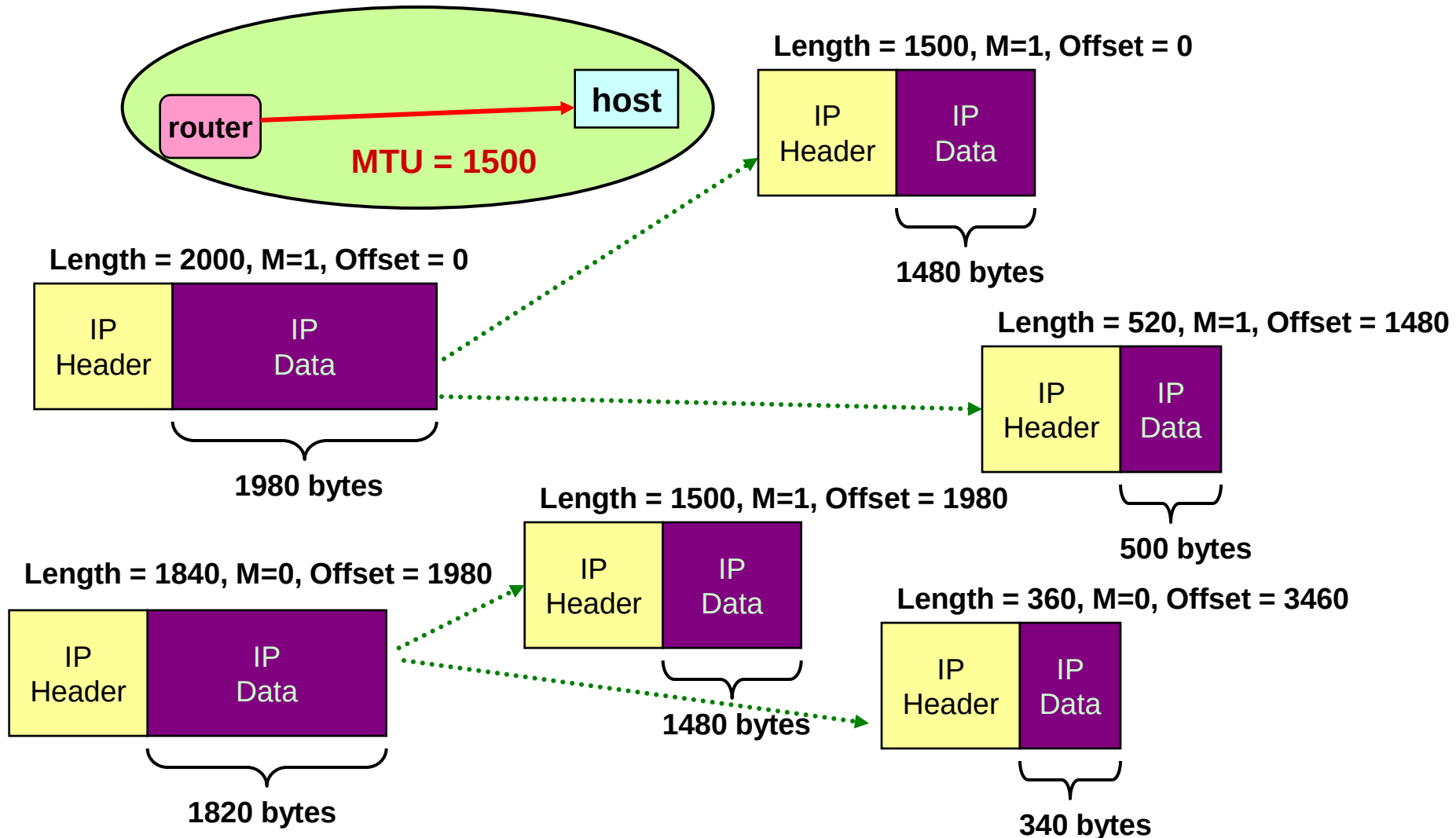
Length = 3820, M=0



IP Fragmentation Step #2



IP Fragmentation Step #3



IP Reassembly

Length = 1500, M=1, Offset = 0



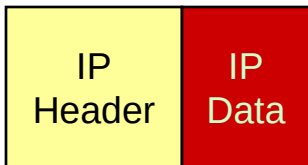
Length = 520, M=1, Offset = 1480



Length = 1500, M=1, Offset = 1980



Length = 360, M=0, Offset = 3460



- Fragments might arrive out-of-order
 - Don't know how much memory required until receive final fragment
- Some fragments may be duplicated
 - Keep only one copy
- Some fragments may never arrive
 - After a while, give up entire process

